Yuanchen Li

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Profile

A junior in computer science from the Honors College at Hangzhou Dianzi University, with strong academic performance and a record of merit-based scholarships. My current research focuses on computer vision and 3D reconstruction, particularly 3D Gaussian Splatting. My broader interests extend to world models for embodied spatial reasoning, and AI for social science—an area I approach from a long-standing immersion in Chinese classical literature, with interests in computational philology, the decipherment of ancient scripts, and the cultural-historical analysis of textual traditions.

Education

Bachelor's Degree in Computer Science, Hangzhou Dianzi University 2023–2027

- **GPA:** 3.3/4 | Selected Coursework: Data Mining (98), Deep Learning (97), Compiler Construction (95), Algorithms and Data Structures (92)

Work and Research Experience

Computer Vision Research Project — Sep 2024 – Mar 2026

- Leading an independent research project that applies 3D Gaussian Splatting (3DGS) to sparse-view CT reconstruction and 3D-consistent multi-organ segmentation, addressing the clinical challenge of reducing radiation dose while maintaining diagnostic-quality volumetric segmentation.
- Developed an SDF-based Gaussian segmentation framework that converts organ masks into signed distance fields and assigns covariance-aware soft labels to 3D Gaussians via trilinear SDF sampling, resulting in improved boundary sensitivity and geometric consistency; the method achieves a mean Dice score of 0.927, substantially outperforming the nnU-Net baseline (0.84).
- **Yuanchen Li**, Tarek S. Elaydi, Haoyu Yang, Minghao Chen*, et al. SOS-3D: SDF-Guided Organ Segmentation with 3D Gaussian Splatting from Sparse-View CT. Under review: *IEEE Journal of Biomedical and Health* Informatics (SCIE; JIF: 6.7; JCR Q1), 2026.

Computer Graphics Project (UC Berkeley CS184) — Jun 2025 – Aug 2025

- Led the final project for Computer Graphics, designing and implementing an end-to-end real-time VR gaming pipeline built on 3D Gaussian Splatting, from raw scene capture to interactive rendering.
- Engineered the full data pipeline: multi-view game-scene capture, camera pose extraction, COLMAP-based Structure-from-Motion calibration, and Gaussian-based 3D scene reconstruction, achieving photorealistic novel-view synthesis.
- Extended the Nerfstudio Splatfacto pipeline to support online scene adaptation by incrementally updating exported Gaussians with newly captured frames in approximately 0.5 seconds, enabling interactive VR rendering of dynamic scene changes without full retraining.
- Ranked 2nd among 54 final project teams evaluated alongside UC Berkeley students, and selected for the UC Berkeley CS184 Showcase.

Honors and Awards

- Outstanding Undergraduate Scholarship, 2023–2024/2024–2025 Academic Year
- Recognized as an OpenHarmony Certified Associate
- Second Prize, Zhejiang College Student Securities Investment Competition (Quantitative Trading Track), 2024

Skills and Interests

- Leadership and teamwork
- Research mentorship and cross-functional collaboration
- **Systems programming:** C, RISC-V Assembly
- PyTorch, CUDA
- Literary creation (works published in Chinese Writers)